

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

The power dissipated by a resistor is given by $P = V^2 / R$, where P is power, V is the voltage across the resistor, and R is the resistance. In this case, the voltage is 12 volts, and the power dissipated is 40 watts. We can rearrange the formula to solve for resistance: $R = V^2 / P = (12)^2 / 40 = 144 / 40 = 3.6$ ohms.

The current drawn from the battery is given by $I = P / V$, where I is current, P is power, and V is voltage. Plugging in the values, we get $I = 40 / 12 = 3.33$ amps.

</think>< answer> E

